Intelligent Candidate Discovery & Ranking

Team FitFinderAi - Methodology & Architecture

1. Core Philosophy & Architecture

To build a true AI system that understands contextual fit while adhering to the strict 5-minute CPU-only constraint, we designed a Two-Phase Hybrid Retrieval Architecture. Keywords are easily gamed, so instead we rely on dense vector embeddings for semantic matching, combined with hard behavioral and trajectory signals.

Phase 1 (Offline): All candidates are embedded using 'all-MiniLM-L6-v2' and stored in a highly optimized FAISS index.
Phase 2 (Online): At runtime, we embed the core job description requirements, perform an instant semantic search to retrieve the Top 10,000 candidates, and apply our behavioral re-ranking algorithm to produce the final Top 100.

2. Constraint Enforcement & Honeypot Detection

Before finalizing scores, our pipeline strictly enforces trajectory and authenticity checks:

- **Honeypot Detection:** We actively filter profiles claiming 'expert' status with 0 months duration, or those whose single job experience duration drastically exceeds their total Years of Experience.
- **Target YoE Bound:** Candidates outside the 4-15 Years of Experience range are filtered out. We grant a slight score boost for candidates inside the 5-9 YoE 'sweet spot' identified in the JD.
- **Strict Red Flags:** Candidates with purely services/consulting backgrounds, pure researchers, or exclusively LangChain API wrappers are strictly disqualified.

3. Hybrid Re-Ranking Scoring

The final candidate score is a weighted combination (0.6 Semantic + 0.4 Behavioral):

• **Semantic Score:** The cosine similarity derived from FAISS, matching the candidate's entire career corpus against the JD embedding.

• **Behavioral Score:** Sourced from Redrob signals. It heavily weights recruiter response rate and interview completion rate, while applying penalties for extended notice periods (>60 days). This ensures we rank capable engineers who are actually available and engaged.

4. Performance & Constraints

By offloading embedding generation to the pre-computation phase, the runtime Stage 3 ranking step operates flawlessly within the 5-minute limit, requiring under 10 seconds and < 500MB RAM to search the FAISS index and score 100K candidates on a single CPU core.